

Understanding Stars

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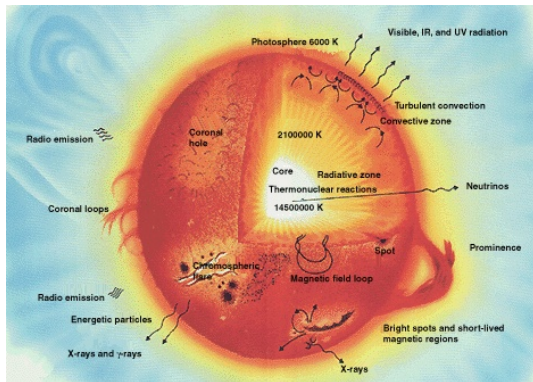
2026

The presentation will provide answers to the following questions:

- What is a star and how does it form?
- How do stars generate energy?
- How do stars enrich the interstellar medium?
- What are the factors governing stellar structure?
- What is a Hertzsprung–Russell diagram and what can we infer from it?
- How do stars evolve over their lifetimes?
- What are the endpoints of stellar evolution?

What is a Star?

A star is a self-gravitating ball of plasma in which hydrostatic equilibrium is maintained between inward gravitational collapse and outward pressure support, sustained over timescales of by nuclear energy release in the core.



What is a Star?

Key observational properties:

- **Luminosity** L (erg s^{-1}): total power radiated.
- **Effective temperature** T_{eff} : defined by $L = 4\pi R^2 \sigma T_{\text{eff}}^4$ (Stefan–Boltzmann).
- **Mass** M : the single most important parameter governing stellar evolution.
- **Radius** R : determined by M , composition, and evolutionary state.
- **Composition**: hydrogen mass fraction X , helium Y , metals Z (where $X + Y + Z = 1$).

The Sun: $M_{\odot} = 1.989 \times 10^{33} \text{ g}$, $R_{\odot} = 6.96 \times 10^{10} \text{ cm}$,
 $L_{\odot} = 3.83 \times 10^{33} \text{ erg s}^{-1}$, $T_{\text{eff},\odot} = 5778 \text{ K}$.

Star Formation

Stars form from the gravitational collapse of cold, dense regions within molecular cloud (giant interstellar structures of H_2 and dust).

A cloud fragment of mass M and temperature T collapses if it exceeds the **Jeans mass**,

$$M_J = \frac{\pi}{6} \rho \lambda_J^3, \quad \lambda_J = \left(\frac{\pi c_s^2}{G \rho} \right)^{1/2}$$

where $c_s = (k_B T / \mu m_H)^{1/2}$ is the isothermal sound speed and μ is the mean molecular weight.

Star Formation

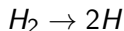
Equivalently, collapse requires that the gravitational energy exceed the thermal energy:

$$|E_{\text{grav}}| = \frac{3GM^2}{5R} > E_{\text{therm}} = \frac{3Mk_B T}{2\mu m_H}$$

This can be used to estimate the age of a star (like our own Sun), known as the **Kelvin–Helmholtz timescale**, under certain assumptions. The age of Sun estimated using this technique comes out to be of the order of 10^7 yr, but other techniques, like Carbon Dating, has contradicting results. This implies there is something else involved as well!

Protostellar Collapse and Pre-Main-Sequence Evolution

Till now, a (sort of a) dispersed cloud of molecular Hydrogen (H_2) condensed into a more compact one, but now the atoms are very close to each other and the collisions cause the H_2 -ball to heat up. Heat can't escape, so this energy goes into dissociating H_2 , i.e.

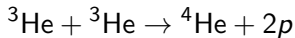
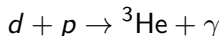
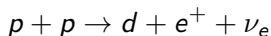


making it a **Protostar**.

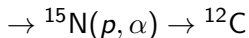
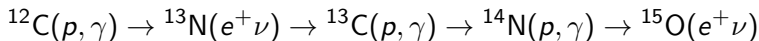
Stellar Energy Generation: Hydrogen Burning

Stars on the main sequence convert hydrogen to helium in the core via two main reaction chains:

- **Proton–proton (pp) chain** (dominant for $M \lesssim 1.3 M_{\odot}$, $T_c \lesssim 1.8 \times 10^7$ K):



- **CNO cycle** (dominant for $M \gtrsim 1.3 M_{\odot}$, $T_c \gtrsim 1.8 \times 10^7$ K):
C, N, O act as catalysts; net result is $4p \rightarrow {}^4\text{He} + 2e^+ + 2\nu_e + 3\gamma$.



Stellar Energy Generation: Hydrogen Burning

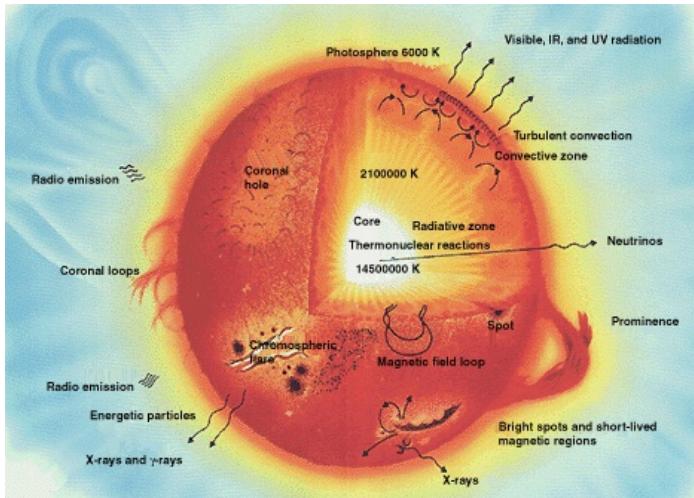


Figure: Star Structure

Stellar Energy Generation: Hydrogen Burning

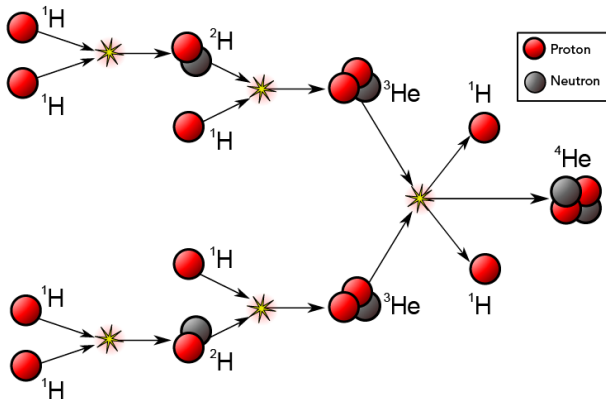
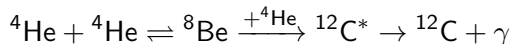


Figure: Hydrogen Burning

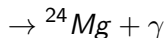
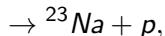
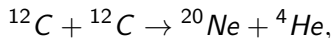
Stellar Energy Generation: Advanced Nuclear Burning

Once the hydrogen in the core is exhausted, more massive stars ignite successive burning stages at progressively higher temperatures:

- **Helium burning** ($T_c \sim 10^8$ K): the triple-alpha process,



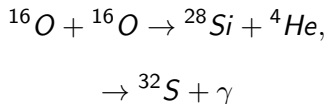
- **Carbon burning** ($T_c \sim 5 \times 10^8$ K):



- **Neon burning** ($T_c \sim 1.5 \times 10^9$ K): photodisintegration of ${}^{20}\text{Ne}$ releases α particles that are recaptured.

Stellar Energy Generation: Advanced Nuclear Burning

- **Oxygen burning** ($T_c \sim 2 \times 10^9$ K):



- **Silicon burning** ($T_c \sim 3 \times 10^9$ K): quasi-statistical nuclear equilibrium produces an iron-peak nucleus (${}^{56}\text{Ni}$, ${}^{56}\text{Fe}$). Iron has the highest binding energy per nucleon making it extremely stable, so no further energy can be extracted by fusion!

Stellar Burning

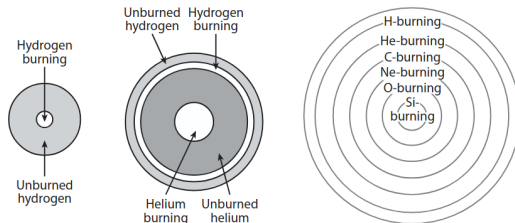


Figure: Illustration 1

Name of process	Fuel	Products	Temperature
Hydrogen-burning	H	He	60×10^6 °K
Helium-burning	He	C, O	200×10^6 °K
Carbon-burning	C	O, Ne, Na, Mg	800×10^6 °K
Neon-burning	Ne	O, Mg	1500×10^6 °K
Oxygen-burning	O	Mg to S	2000×10^6 °K
Silicon-burning	Mg to S	Elements near Fe	3000×10^6 °K

Figure: Illustration 2

Figure: Stellar Burning.

Stellar Burning

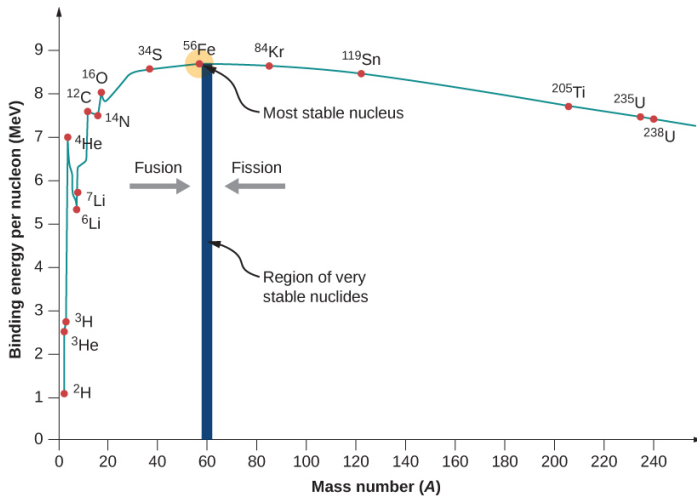


Figure: Binding energy per nucleon.

Stellar Nucleosynthesis and Chemical Enrichment

Stars are the primary factories of heavy elements in the Universe.
The main production sites are:

- Big Bang nucleosynthesis: H, D, ^3He , ^4He , Li; but nothing heavier than Be.

Stellar Nucleosynthesis and Chemical Enrichment

- **Stellar nucleosynthesis:**

- *CNO, Ne–Ti*: hydrostatic burning in massive stars.
- *Iron peak*: explosive nucleosynthesis in CCSN and SN Ia.
- *s-process* (slow neutron capture): in Redgiants; produces Sr, Ba, Pb.
- *r-process* (rapid neutron capture): in NS–NS mergers and possibly CCSNe; produces Au, Pt, Eu, Th, U.
- *p-process*: proton capture / photodisintegration in SN shocks; produces proton-rich isotopes.

- **Cosmic-ray spallation**: Li, Be, B produced by CR interactions with CNO nuclei in the ISM.

Stellar Populations: Overview

- Classification based on metallicity, age, and galactic location
- Reflects chemical evolution of the universe over cosmic time
- Three broad categories: Population I, II, III
- Metallicity affects opacity $\rightarrow$ directly impacts stellar structure (radius, T_{eff} , HR diagram position)

Population I

- Metal-rich: $Z \sim 0.02$ (solar-like or higher)
- Young to intermediate age
- Found in galactic disk, spiral arms, open clusters
- Example: the Sun
- Kinematically “cool” – circular, low-inclination orbits

Population II

- Metal-poor: $Z \sim 10^{-3}$ – 10^{-2} solar or lower
- Old stars ($\gtrsim 10$ Gyr)
- Found in galactic halo, globular clusters, bulge
- Formed early, before significant chemical enrichment
- Kinematically “hot” – eccentric, randomly oriented orbits

Population III

- First generation of stars, formed shortly after the Big Bang
- Essentially zero metallicity ($Z \approx 0$): primordial H, He, trace Li
- Theorized to be very massive (tens–hundreds of $M_{\odot}$)
 - No metal-line cooling $\rightarrow$ inefficient gas fragmentation $\rightarrow$ massive stars
- Drove cosmic reionization; produced first heavy elements
- Not yet directly observed – inferred from extremely metal-poor stars and high- z surveys (e.g., JWST)

Chemical Enrichment: III $\rightarrow$ II $\rightarrow$ I

- Pop III stars die (often as supernovae) $\rightarrow$ seed ISM with first metals
- Pop II stars form from this enriched gas, still metal-poor
- Continued enrichment leads to metal-rich Pop I stars (like the Sun)
- Each generation's structure differs due to opacity changes:
 - Higher $Z \rightarrow$ higher opacity $\rightarrow$ cooler, more extended envelopes

Modes of Heat Transfer



Figure: Modes of Heat Transfer.

Modes of Heat Transfer

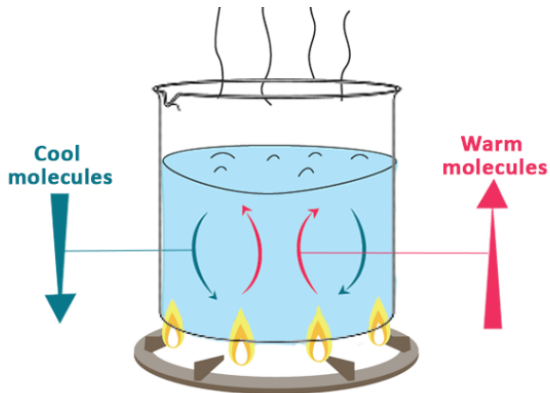


Figure: Modes of Heat Transfer.

Modes of Heat Transfer

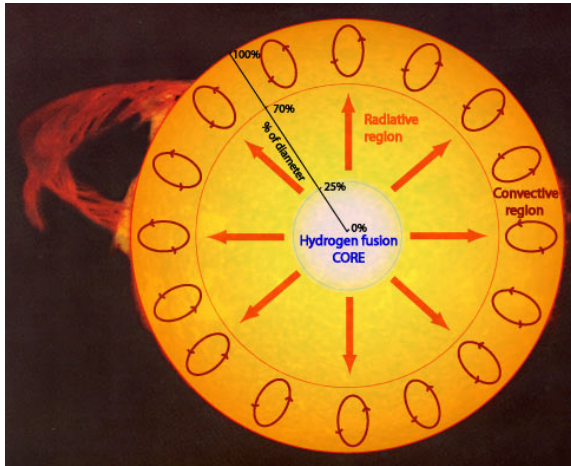


Figure: Modes of Heat Transfer in stars.

Equations of Stellar Structure

A spherically symmetric star in hydrostatic equilibrium is governed by four coupled differential equations (Eddington's stellar structure equations):

- **Mass continuity:**

$$\frac{dM(r)}{dr} = 4\pi r^2 \rho(r)$$

- **Hydrostatic equilibrium:**

$$\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

Equations of Stellar Structure

- **Energy generation:**

$$\frac{dL(r)}{dr} = 4\pi r^2 \rho(r) \varepsilon$$

where ε ($\text{erg g}^{-1} \text{s}^{-1}$) is the local energy generation rate.

Some contributions to ε ,

- Nuclear fusion (pp-chain, CNO cycle, triple-alpha)
- Gravitational contraction (pre-main-sequence, late stages)
- Strong temperature dependence: $\varepsilon \propto T^n$ (n large for CNO)

Equations of Stellar Structure

- **Energy transport** (radiative):

$$\frac{dT}{dr} = -\frac{3\kappa\rho(r)L(r)}{64\pi\sigma T^3 r^2} \quad ; \text{ where } \kappa \text{ is the opacity (cm}^2 \text{ g}^{-1}\text{)}$$

Some contributions to κ ,

- Electron scattering (Thomson scattering)
- Bound-bound transitions or bound-free/free-free absorption (line opacity)

Kramers' law (approx): $\kappa \propto \rho T^{-7/2}$

Polytropic Models

- Simplified models using $P = K\rho^{1+1/n}$
- Reduces structure equations to the Lane-Emden equation,

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n$$

where we used $\rho = \rho_c \theta^n$, $r = \alpha \xi$, $\alpha^2 = (n+1)K\rho_c^{1/n-1}/4\pi G$.

- Useful approximations,
 - $n = 3/2$: non-relativistic degenerate gas (white dwarfs)
 - $n = 3$: radiation-dominated / relativistic degenerate gas
 - $n = 0$: uniform density

Stellar Magnetic Fields: Origin

- Generated by dynamo action: interplay of convection, rotation, and differential rotation
- Requires:
 - Convective zone (turbulent plasma motions)
 - Rotation (Coriolis effect organizes flows)
 - Differential rotation (shearing of field lines)
- Magnetohydrodynamics (MHD) governs field evolution:

$$\frac{\partial \vec{B}}{\partial t} = \nabla \times (\vec{v} \times \vec{B}) + \eta \nabla^2 \vec{B}$$

Magnetic Field Strengths

- Sun: average surface field $\sim 1\text{--}2$ G; sunspots up to ~ 3000 G
- Magnetic Ap/Bp stars: $10^2\text{--}10^4$ G (fossil fields)
- White dwarfs: up to 10^8 G
- Neutron stars: $10^{12}\text{--}10^{13}$ G typical; magnetars up to 10^{15} G
- Field strength scales with stellar type and evolutionary state

Magnetic Pressure and Plasma Beta

- Magnetic pressure:

$$P_B = \frac{B^2}{8\pi}$$

- Plasma beta parameter:

$$\beta = \frac{P_{gas}}{P_B}$$

- $\beta \gg 1$: gas pressure dominates (stellar interior)
- $\beta \lesssim 1$: magnetic pressure dominates (sunspots, corona)
- Determines how magnetic fields influence local structure and dynamics

Magnetic Effects on Structure

- Magnetic pressure modifies hydrostatic balance locally:

$$\frac{d}{dr} \left(P + \frac{B^2}{8\pi} \right) = - \frac{Gm\rho}{r^2}$$

- Suppresses convection locally (sunspots are cooler, darker)
- Can support extended structures against gravity (magnetic flux tubes)
- In magnetars, magnetic pressure can be comparable to/exceed matter pressure, deforming the star

Magnetic Flux Tubes and Sunspots

- Strong, concentrated magnetic flux emerges through the photosphere
- Sunspots: regions of strong vertical field ($\sim \text{kG}$) where convection is inhibited
- Cooler than surroundings ($T \sim 4000 \text{ K}$ vs $\sim 5800 \text{ K}$) – appear dark
- Flux tube structure maintained by magnetic tension balancing buoyancy and pressure

Fossil Fields vs Dynamo Fields

- **Fossil fields:** remnants of field acquired during star formation, frozen into radiative regions (Ap/Bp stars, white dwarfs)
- **Dynamo fields:** continuously regenerated by convection + rotation (Sun-like stars)
- Different field morphologies and decay timescales
- Fossil fields can be very stable (large-scale, ordered); dynamo fields vary cyclically (e.g., 11-year solar cycle)

Rotational Effects on Structure

- Rotation introduces centrifugal force, modifying effective gravity
- Rapid rotators become oblate (equatorial radius $>$ polar radius)
- Von Zeipel effect: gravity darkening – poles hotter/brighter than equator in rapid rotators
- Differential rotation (varying Ω with radius/latitude) is key for dynamo action
- Coupling between rotation and magnetic field $\rightarrow$ angular momentum transport (magnetic braking)

Stellar Atmospheres and Spectral Classification

The stellar spectrum encodes the photospheric temperature, composition, pressure (gravity), etc. The Harvard classification organises stars by decreasing T_{eff} :

Class	T_{eff} (K)	Colour	Characteristic lines
O	$> 30,000$	blue-white	He II, N III, O III ionised
B	10,000–30,000	blue-white	He I, H Balmer
A	7,500–10,000	white	H Balmer strongest
F	6,000–7,500	yellow-white	Ca II H&K, metals
G	5,200–6,000	yellow	Ca II H&K, Fe, CN
K	3,700–5,200	orange	Ca II, TiO begins
M	$< 3,700$	red	TiO, H ₂ O bands

The mnemonic: *Oh Be A Fine Girl/Guy, Kiss Me!*

Stellar Atmospheres and Spectral Classification

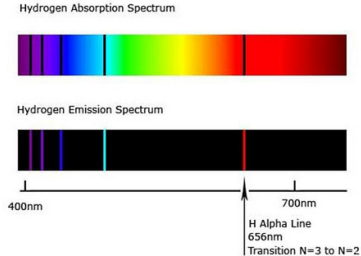


Figure: Illustration 1

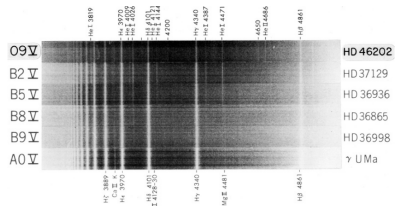


Figure: Illustration 2

Figure: Spectrum.

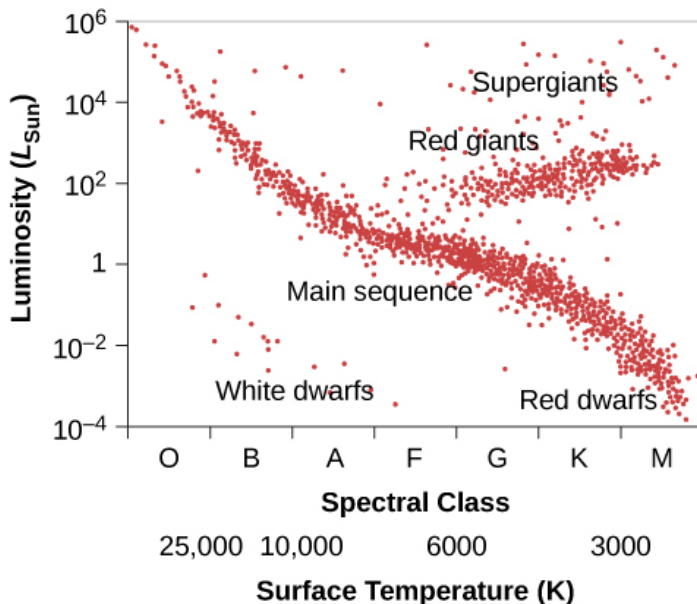
The Hertzsprung–Russell Diagram

The Hertzsprung–Russell (H–R) diagram plots stellar luminosity L vs. effective temperature T_{eff} (or equivalently, absolute magnitude vs. colour index). It is the central organising diagram of stellar astrophysics.

Major features:

- **Main sequence (MS)**: a diagonal band from hot, luminous O stars to cool, dim M dwarfs. Stars on the MS are burning H in their cores.
- **Giant branch**: luminous, cool stars ($T_{\text{eff}} \sim 4000\text{--}5000\text{ K}$) with expanded envelopes.
- **Horizontal branch (HB)**: core He-burning at roughly constant $L \sim 50 L_{\odot}$.
- **White dwarf sequence**: faint, hot, compact remnants cooling at constant radius.
- **Instability strip**: vertical band containing pulsating variables (Cepheids, RR Lyrae).

The Hertzsprung–Russell Diagram



The Hertzsprung–Russell Diagram

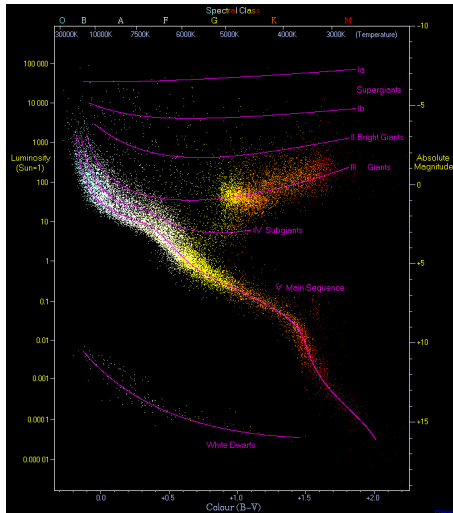


Figure: HR diagram.

The Hertzsprung–Russell Diagram

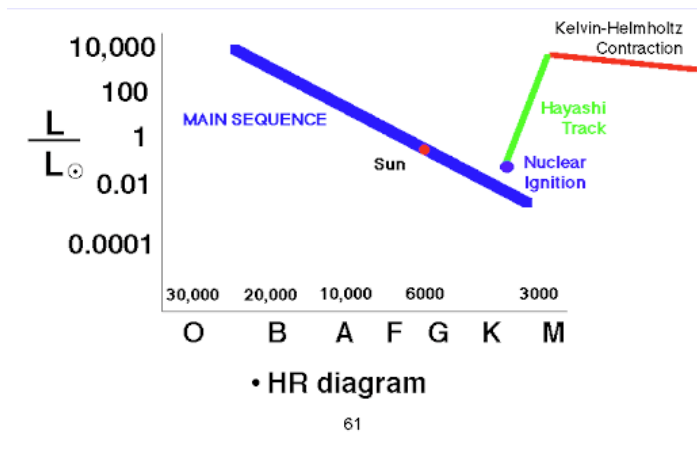


Figure: HR diagram.

Low- and Intermediate-Mass Stellar Evolution ($M \lesssim 8 M_{\odot}$)

Following H exhaustion in the core, the evolutionary sequence is:

- **Subgiant branch:** H-exhausted core contracts and heats; H-burning shell ignites; envelope expands and cools.
- **Red Giant Branch (RGB):** star ascends the giant branch at roughly constant T_{eff} ; deepening convective envelope (first dredge-up) brings CN-cycle products to the surface.
- **Helium flash:** for $M \lesssim 2.3 M_{\odot}$, the degenerate He core ignites in a violent, off-centre thermal runaway at $L_{\text{He}} \sim 10^{11} L_{\odot}$ (over seconds). Electron degeneracy is lifted; star settles onto the horizontal branch.

Low- and Intermediate-Mass Stellar Evolution ($M \lesssim 8 M_{\odot}$)

Following H exhaustion in the core, the evolutionary sequence is:

- **Horizontal Branch (HB):** core He burning + shell H burning. Duration $\sim 10^8$ yr.
- **Asymptotic Giant Branch (AGB):** He and H burning shells alternate (thermal pulses). Third dredge-up brings ^{12}C to the surface; star may become carbon-rich (C star).
- **Mass loss:** strong AGB winds ($\dot{M} \sim 10^{-5} - 10^{-4} M_{\odot} \text{ yr}^{-1}$) eject the envelope, producing a planetary nebula and leaving behind a white dwarf.

High-Mass Stellar Evolution ($M \gtrsim 8 M_{\odot}$)

Massive stars evolve rapidly and end their lives explosively:

- **Main sequence:** CNO-cycle burning in a large convective core; strong radiation-driven winds ($\dot{M} \sim 10^{-6} - 10^{-5} M_{\odot} \text{ yr}^{-1}$) strip mass progressively.
- **Post-MS:** star expands to a red supergiant (RSG, $T_{\text{eff}} \sim 3500 \text{ K}$, $R \sim 10^3 R_{\odot}$) or, for the most massive, an LBV (Luminous Blue Variable) or Wolf-Rayet (WR) star (stripped envelope, bare He or C/O core).
- **Onion-shell structure:** successive burning stages (He, C, Ne, O, Si) occur in nested shells on rapidly decreasing timescales. Silicon burning lasts only ~ 1 day.

High-Mass Stellar Evolution ($M \gtrsim 8 M_{\odot}$)

Massive stars evolve rapidly and end their lives explosively:

- **Iron core:** once an $\sim 1.4 M_{\odot}$ (M_{Ch}) iron core builds up, electron degeneracy pressure can no longer support it. Gravitational collapse begins.
- **Core-collapse supernova (CCSN):** the inner core reaches nuclear density ($\rho \sim 3 \times 10^{14} \text{ g cm}^{-3}$) and bounces. A shock wave propagates outward, revived by $\sim 10^{53}$ erg of neutrino energy deposition, and unbinds the stellar envelope.

Summary

- Stars are self-gravitating plasma spheres sustained by nuclear fusion, governed by four coupled differential equations of stellar structure: mass continuity, hydrostatic equilibrium, energy generation, and energy transport.
- The mass of a star is the dominant parameter. It determines luminosity ($L \propto M^4$), lifetime ($t_{\text{MS}} \propto M^{-2.5}$), and final fate (WD, NS, or BH).
- The H–R diagram organises all evolutionary stages; spectral classification (OBAFGKM) links observed colours and lines to T_{eff} and $\log g$.

Summary

- Low-mass stars die as planetary nebulae + white dwarfs; massive stars undergo core-collapse supernovae, leaving neutron stars or black holes.
- Stars are the engines of cosmic chemical evolution: from primordial H and He, they have built up the periodic table through hydrostatic and explosive nucleosynthesis, s-process, and r-process.

White Dwarfs

A **white dwarf** (WD) is the degenerate remnant left after the initial burning phase. The pressure at the core a White Dwarf isn't enough to sustain a nuclear fusion reaction.

Structure:

- Supported by non-relativistic electron degeneracy pressure.
- Composition: C/O core (from He burning), thin He and H envelopes.

Chandrasekhar mass limit: The star is unstable and accretion beyond M_{Ch} in a binary triggers a **Type Ia supernova**.

WDs simply cool over time, the WD luminosity function constrains the age of the Galactic disk.

Type Ia Supernovae

A **Type Ia supernova** (SN Ia) is the thermonuclear explosion of a C/O white dwarf accreting mass from a binary companion or merging with another WD,

- Carbon ignites in the degenerate core at $M \rightarrow M_{\text{Ch}}$; a deflagration/detonation (DDT model) incinerates the star.
- Peak luminosity: $L_{\text{peak}} \approx 10^{43} \text{ erg s}^{-1}$ ($M_V \approx -19.3$), powered by the decay chain $^{56}\text{Ni} \rightarrow ^{56}\text{Co} \rightarrow ^{56}\text{Fe}$.

The 1998 discovery of cosmic acceleration via SN Ia was awarded the 2011 Nobel Prize in Physics.

Neutron Stars and Pulsars

The remnant of a Core-Collapse Supernova (CCSN) (for progenitors $8 \lesssim M/M_{\odot} \lesssim 20$) is a neutron star (NS), supported by neutron degeneracy pressure and nuclear forces. The density of a Neutron Star is like compressing an entire mountain to the size of a teaspoon!

Pulsars: rapidly rotating NS with strong magnetic fields ($B \sim 10^{12}\text{--}10^{13}$ G). The rotating dipole emits beamed electromagnetic radiation; if the beam sweeps past Earth, we observe regular pulses.

Pulsars

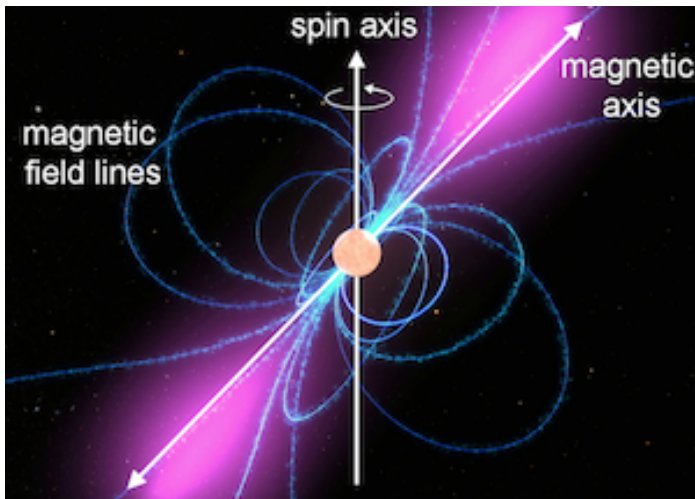


Figure: Pulsars

Pulsars

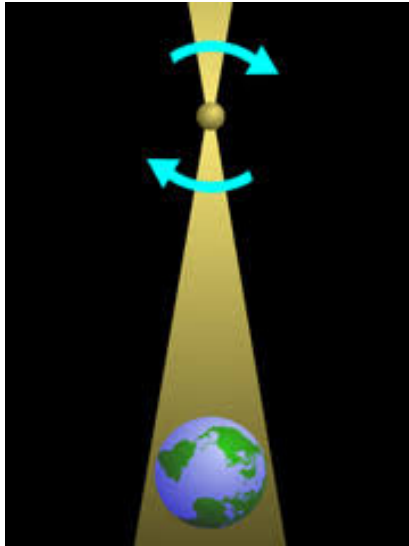


Figure: Detection of Pulsars. □ ▶

Pulsars

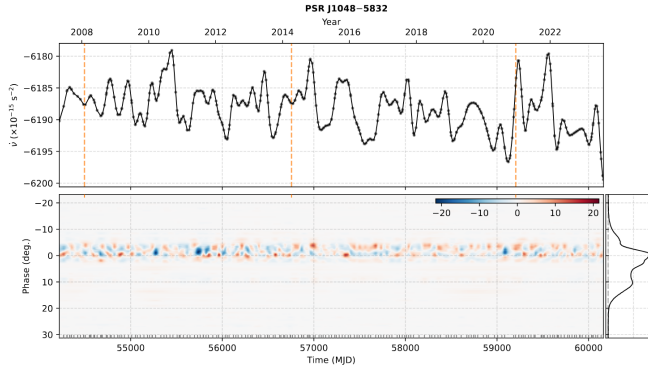


Figure: Variability.

Stellar Variability: Overview

- Variable stars: brightness changes over time due to intrinsic or extrinsic causes
- **Intrinsic:** physical changes in the star itself (pulsation, eruption)
- **Extrinsic:** geometric effects (eclipses, rotation with spots)
- Variability provides a powerful probe of stellar interiors (asteroseismology)

Pulsating Variables

- Periodic expansion/contraction $\rightarrow$ changes in L and T_{eff}
- **Cepheids**: classical pulsators; period-luminosity relation $\rightarrow$ distance indicators
- **RR Lyrae**: older, lower-mass; found in globular clusters; standard candles
- **Mira variables**: long-period (100–1000 days), AGB stars
- **Delta Scuti, ZZ Ceti (pulsating WDs)**: other important classes

$$P\sqrt{\rho} \approx \text{constant} \quad (\text{period-mean density relation}) \quad (1)$$

Black Holes

For very massive progenitors ($M \gtrsim 20\text{--}25 M_{\odot}$, depending on metallicity and mass loss), the remnant mass exceeds the maximum NS mass and a black hole forms.

A non-rotating BH is characterised solely by its mass M (Schwarzschild metric):

$$ds^2 = - \left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

with Schwarzschild radius $r_s = 2GM/c^2 \approx 3 (M/M_{\odot})$ km.



Figure: M87

Black Holes

- When a black hole forms, or when two black holes violently collide, the resulting event horizon is temporarily warped and distorted.
- To settle down into a perfect, stable sphere, the black hole "rings" like a struck bell, radiating that excess energy away as ripples in the fabric of spacetime.
- Quasi-Normal Modes (QNMs) are the specific frequencies of this fading ringdown signal act like a unique fingerprint, allowing physicists to determine the exact mass and spin of the black hole!

Thank You!

Questions?